

Exercise solutions for chapter-4

Hands ON ML

1. If the training set has millions of features , Gradient Descent is the better choice , as normal equation scales poorly with large data, although it also depends on our use case and also no.of training instances to decide further
2. Gradient Descent algorithms would be affected due to elongated bowl scenario, normal forms are not affected by the scales of the data , We can scale the features using standard scaler
3. Gradient Descent cannot stay stuck in local minima in logistic regression model , as the cost function is a convex function , there is only global minima
4. Yes, although some converge faster than others , depending on the learning rate , if the optimization problems is convex and learning rate is not too big , it will converge , although if learning rate is too high or in cases of stochastic GD and mini-batch GD , if learning rate is not reduced it wont converge
5. If validation error is constantly going up , one of two things happening , if training error is reducing and validation error is going up , it is overfitting the data , but if the training error is increasing and so is the validation error , it can mean learning rate is too high and it is diverging
6. Due to random nature of stochastic GD , and mini-batch GD , the increase of validation error doesnt tell the full story ,stopping them early can lead to not have the optimum value , one thing that can be done is to save the model at regular intervals , and if the validation error is going up for say 5 folds continuously then load the previous model
7. Highly depends , both stochastic and mini-batch are random in nature , until and unless the traning rate is reduced substantially after some iterations , it will continue to bounce around , only batch GD is the one , that always converges given a good learning rate and sufficient epochs
8. Model is overfitting , in case of polynomial model , either reduce the degress of curve fitting or add regularization like l1 and l2

9. If both errors are high , it means the model is underfitting the data , in this case it is high bias , and we should reduce the regularization constraint
10. We would want to use ridge regression instead of plain one , when data points are less , and we are willing to compromise a little bias in the training set for long term - lower variance in the test set Lasso Regression uses an ℓ_1 penalty, which tends to push the weights down to exactly zero. This leads to sparse models, where all weights are zero except for the most important weights. This is a way to perform feature selection automatically, which is good if you suspect that only a few features actually matter. When you are not sure, you should prefer Ridge Regression. Elastic Net is generally preferred over Lasso since Lasso may behave erratically in some cases (when several features are strongly correlated or when there are more features than training instances). However, it does add an extra hyperparameter to tune. If you want Lasso without the erratic behavior, you can just use Elastic Net with an `l1_ratio` close to 1.
11. Since the classes are not exclusive , i.e. combinations are possible e.g. it can be outdoor and day , it can also be outdoor and night , so two logistic models are preferred , as softmax assumes that the outcomes are mutually exclusive which is not the case here